MATHEMATICAL PHYSICS

Using AI, Mathematicians Find Hidden Glitches in Fluid Equations

By CHARLIE WOOD

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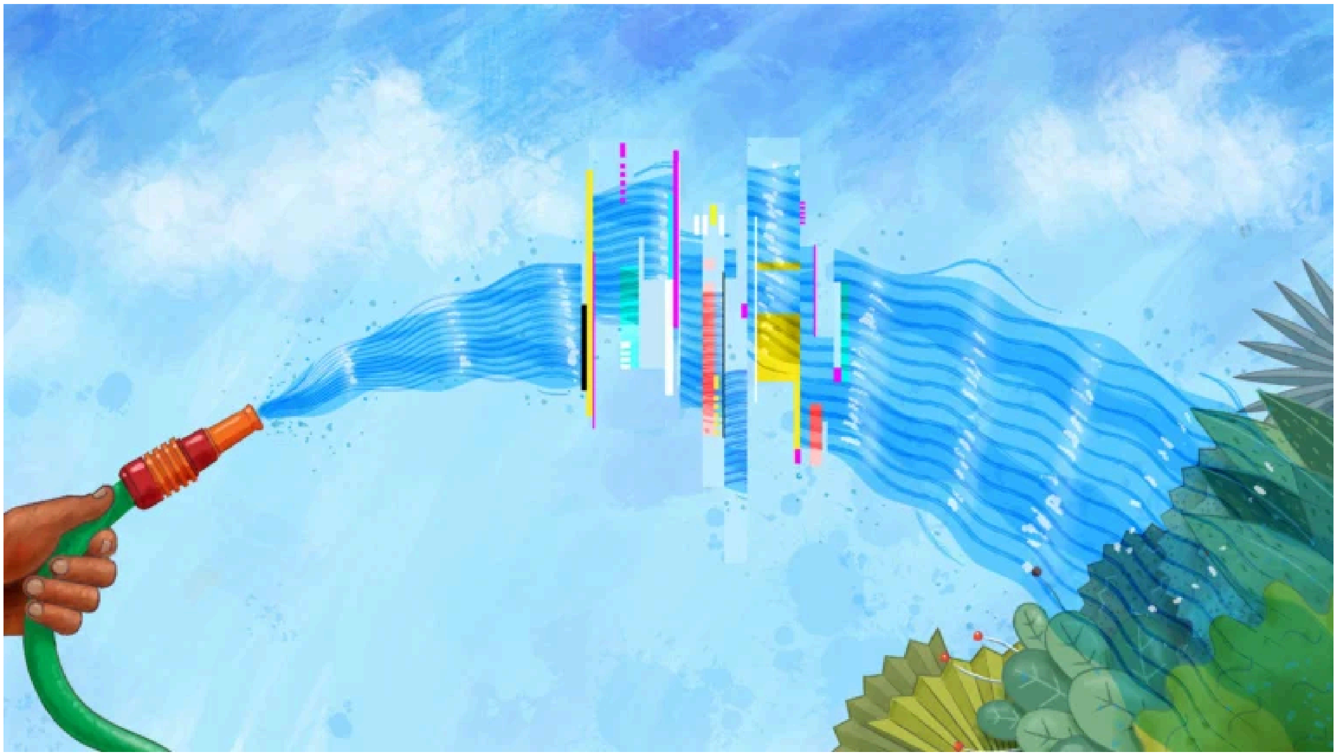
A $1 million prize awaits anyone who can show where the math of fluid flow breaks down. With specially trained AI systems, researchers have found a slew of new candidates in simpler versions of the problem.



as an unimpeachable theory of how fluids in the real world behave — from ocean currents threading their way between the continents to air wrapping around an aircraft's wings.

Nevertheless, many mathematicians suspect that glitches hide deep within the equations. They have a hunch that in certain situations, the theory fails. In these cases, the equations will predict a fluid moving in some unphysical, incomprehensible way — spinning into an impossibly fast vortex, for instance, or instantly blow up — would come with a $1 million reward. And so, as a prelude to solving the Navier-Stokes problem, mathematicians have searched for blowups (also called singularities) in an assortment of simplified fluid equations, such as those that operate in only one dimension.

They've found them. But essentially all the singularities they've identified have been "stable," meaning that they can form in many possible ways. In the most realistic fluid theories, including Navier-Stokes, blowups (if unstable, so we never see them," said Charlie Fefferman, a mathematician at Princeton University who formulated the million-dollar Navier-Stokes challenge.

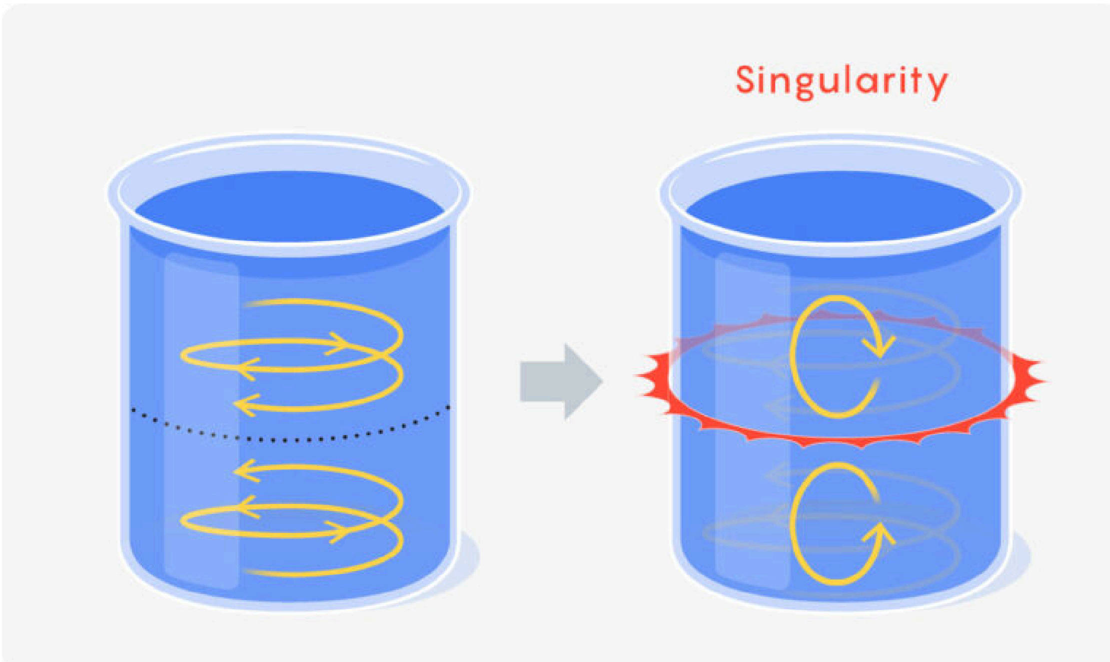
Now one group of mathematicians has developed a way of training machines to spot these phantom glitches. In a preprint posted in September, they reexamined simpler fluid equations already known to host a stable singularity. There, they found additional potential blowup scenarios — including unstable to rigorously prove that the ones they have found do indeed blow up. But their success in uncovering potential unstable singularities in simple models raises hopes that it will also be possible to find unstable blowups in higher-stakes scenarios.

"The idea of an unstable singularity no longer prevents the discovery of the singularity," said Fefferman, who was not involved in the new research.

In a more complicated setup, gentle currents might merge into whirlpools and eddies. The great mystery is whether every solution — every single possible fluid history that satisfies the Navier-Stokes equations — makes sense everywhere and always.

But tackling the Navier-Stokes equations for fluids in three dimensions is unspeakably difficult, so mathematicians have started with easier versions of the problem. For instance, the Euler equations assume that fluids flow with no use a computer to simulate the fluid’s motion and get an approximate sense of the conditions that seem to produce a blowup. If you can narrowly identify the blowup-producing conditions, you might be able to use that knowledge to rigorously prove that a blowup truly exists.

That’s the approach that Thomas Hou and Guo Luo took in 2013, when they simulated a digital liquid in a can. They set the top half of the liquid spinning in one direction and the bottom half in much the liquid spins around a point) got big — bigger than their computer could handle.



It took Hou and another collaborator, Jiajie Chen, nearly a decade to remove the "alleged." In 2022, they used a computer to prove that the singularity candidate implied the existence of a true singularity. It was a landmark proof, and it got mathematicians hungry to push the frontier even further.

The research depended on computer simulations, which meant that tiny adjustments to the initial state of the digital fluid (or any digital rounding errors) wouldn't affect the to that initial arrangement, no matter how small, would prevent the fluid from blowing up.

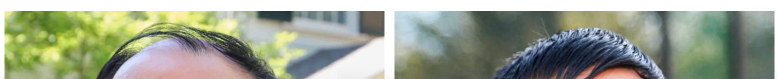
Many mathematicians conjecture that if singularities do lurk in more realistic fluid equations, they'll be unstable like this, springing up without warning.

They'll also be far harder to find.

Going Finite

have to evolve the fluid flawlessly from one moment to the next, since even the smallest deviation will tip it onto a path that doesn't blow up.

Computers aren't capable of infinite precision. They'll inevitably introduce numerical errors that, though tiny, will stop the unstable singularity from forming. "It's like the wind blowing on your pen," Buckmaster said.



from Thomas Hou (left) and Jiajie Chen in 2022.

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Vicki Chiu; Courtesy of Jiajie Chen

As a result, almost all blowup candidates have been stable.

So Buckmaster and his colleague Ching-Yao Lai, now at Stanford University, began to work out a potentially wind-proof way of finding unstable ones.

highly efficient “training” process of guessing, checking, and refining until the function can perform some desired task. For instance, if you calibrate a neural network using thousands of labeled photos of cats and dogs, it “learns” to take in unlabeled images it’s never seen before and label them “cat” or “dog.”

Buckmaster and Lai turned to what’s known as a physics-informed neural network, or PINN. Unlike an image-classifying neural network, a PINN doesn’t learn by studying external data.

But no computer technique can directly render the infinite nature of a singularity. Imagine playing the simulation of your fluid and watching it move forward in time. You might represent some quantity, such as the velocity at different points in the fluid, as a curve on a graph. As the fluid changes over time, you'll see that curve change as well, like a movie. If the curve gets much steeper from one frame to the next, the fluid might be approaching a singularity. The simulation, however, can't reach that final destination. The computer will called self-similarity. This means that there is a way of stretching the velocity curve in one frame to match the steeper velocity curve in a later frame. And so, if you want to catch a potential singularity, you no longer need to try to watch the curve get infinitely steep. Instead, you can zoom in on the steepening section of the curve while the movie plays in a way that neutralizes the steepening. From this new, dynamic perspective, the curve gets closer and closer to a frozen curve of finite steepness instead. This transformation renders the target — the frozen unique zoom rate that makes a singular solution appear frozen and finite.

At first, their PINN turned up only known candidates. In 2022, for instance, Buckmaster and Lai, along with Lai's postdoctoral researcher Yongji Wang and Javier Gómez-Serrano of Brown University, used a PINN to home in on the stable blowup that Hou and Luo had found in 2013. (Hou and Chen would prove its existence later that year.)

neural networks to look for extremely subtle ways of making fluid equations glitch.

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Dan Komoda/ Institute for Advanced Study; Jason Rossi/Brown University

They also rediscovered a known singularity candidate in the Córdoba-Córdoba-Fontelos (CCF) equations, which describe a simpler one-dimensional fluid. That singularity candidate stepping a fluid forward in time. Rather, it went after the frozen limit directly.

"There's no time, so you don't care that it's unstable," Buckmaster said. "You just try to solve the [equation] itself."

A New Stable of Instability

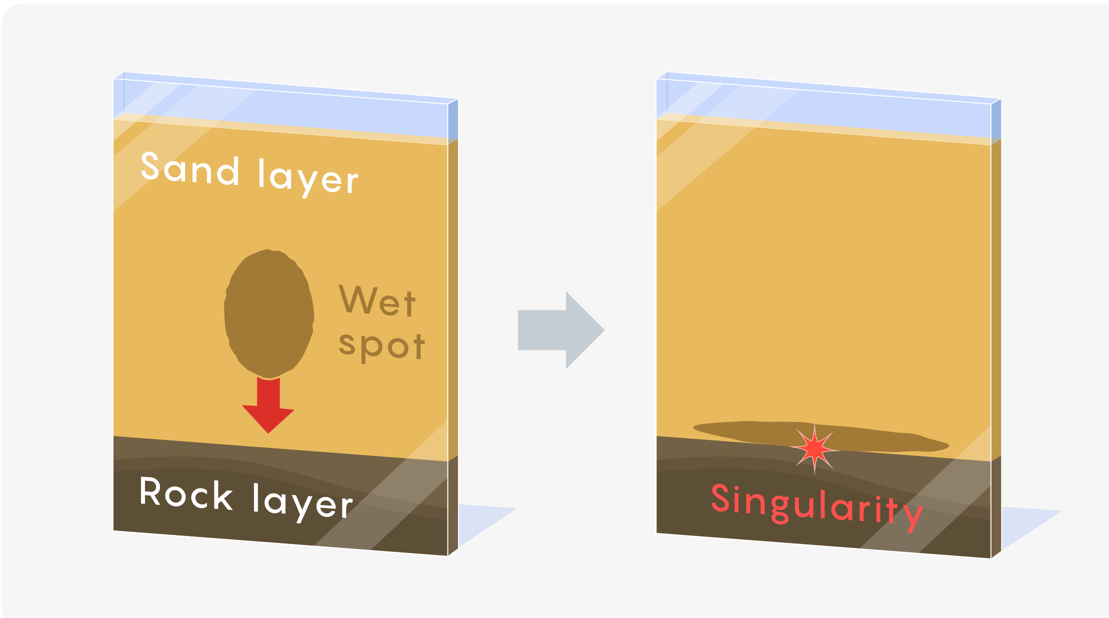
The mathematicians were excited by the prospect of using their PINN to find new unstable singularity candidates. They teamed up further tuned the structure of the PINNs to guide them toward solutions with features they knew the singularities should have.

As they did so, their PINNs got better at spotting singularity candidates. A lot better.

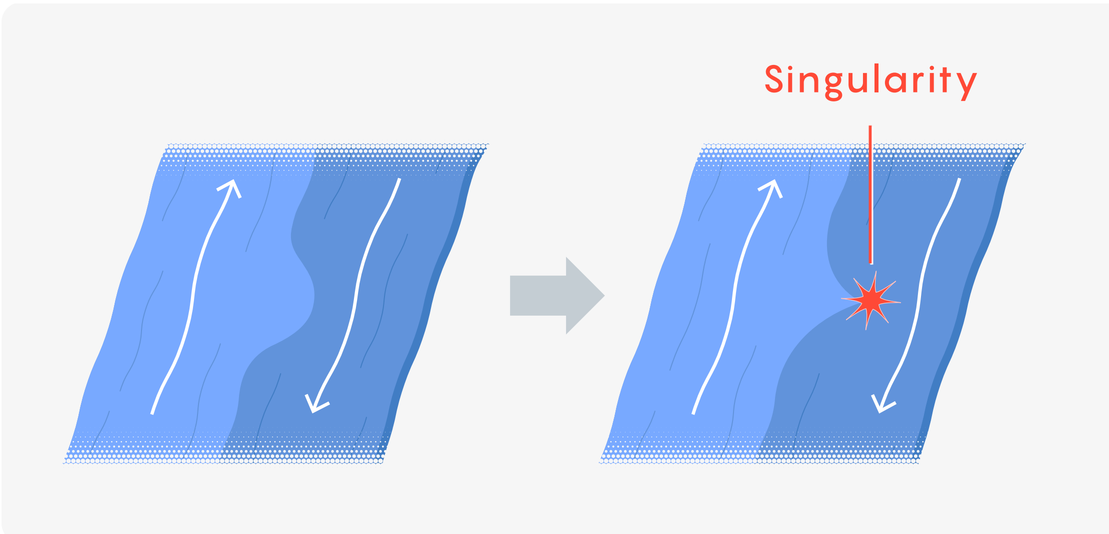
In September, their collaboration of more than 20 researchers unveiled a host of singularities that had never been seen before, most of them unstable.

even more easily when the setup was tweaked in subtle ways.

They also looked at equations describing how a fluid filters through an incompressible porous medium, such as soil or rock, in two dimensions. No one had ever found singularity candidates in this setup. They found four — one stable, three unstable. All involved a similar setup that can be visualized in a thought experiment, although in reality no scientist would be able to adjust the fluid with the endless it smacks into the rock layer, and a property related to the fluid's density seems to blow up.



Finally, the team returned to the oneshaped kink into this interface, it sharpens into a singular cusp.



Notably, like the Navier-Stokes equations (and unlike the other two kinds of equations the

“We are trying to isolate the technical difficulties one by one,” Gómez-Serrano said.

Crucially, none of these new singularity candidates has been proved. But Gómez-Serrano expects that they can be, because the PINN’s approximations are so precise. And the more precise a candidate is, the easier it is to prove that it’s a true singularity. Compared to when the group first unleashed their PINN a few years ago, they’ve gotten about a billion times more accurate.

Race To Escape the Boundary

Yongji Wang, a researcher at Google Deepmind, developed many of the technical innovations that made the discovery of new unstable singularity candidates possible.

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Sang He

The million-dollar question, or technically the warm-up question to the million-dollar question, is whether the DeepMind collaboration can now use their PINN machinery to find a

"You're building a robust tool to find things that are very hard to find," Buckmaster said.

Other mathematicians, however, point out that past performance does not guarantee future returns, because an unbounded fluid is nothing like a bounded one. "It's a completely different animal," said Diego Córdoba, a mathematician at the Institute of Mathematical Sciences in Spain and one of the Córdobas of the CCF model. (His father is the other.)

getting their approach to work for a boundary-free Euler fluid. (Córdoba had fretted that the DeepMind collaboration was about to beat them to that goal, but to his relief, their PINNs aren't yet powerful enough to crack the problem. The solutions they've found, he said, "are unstable, but not that unstable.")

Another competitor, Tarek Elgindi of the University of California, San Diego, has already had success working in boundary-free settings (with other caveats) and is intent on extending

If someone can, then it will be on to Navier-Stokes. But despite the recent rash of progress in finding new fluid glitches, mathematicians hesitate to raise their hopes too high.

"You may daydream, but only for a day or two," Gómez-Serrano said. "You don't have good enough ideas. Then the daydream stops."

Clarification: January 13, 2026
This article has been updated to reflect Ching-Yao Lai's contribution to the development of the